

# TM-DeepNet: A Hybrid Deep Learning Model With Explainable AI for Acute Lymphoblastic Leukemia Stage Classification

Tahmid Enam Shrestha<sup>1</sup>, Mah-Isha Khan<sup>2</sup>, Razuan Karim<sup>3</sup>

<sup>1,2,3</sup>Department of Computer Science, American International University-Bangladesh  
Dhaka, 1229, Bangladesh

Email: <sup>1</sup>25-93790-2@student.aiub.edu, <sup>2</sup>25-93788-2@student.aiub.edu, <sup>3</sup>rkarim@aiub.edu

**Abstract**—Acute Lymphoblastic Leukaemia (ALL) is an aggressive form of blood cancer that originates in the bone marrow and spreads into the bloodstream, which has been a deadly illness for patients. Timely diagnosis is essential for successful treatment and improved recovery outcomes. While several studies have investigated ALL stage detection using deep learning techniques, research gaps remain in terms of achieving higher accuracy, reliance on a single small dataset, and the lack of explainable AI techniques to ensure transparency in decision-making. This study introduces a robust hybrid deep learning model, TM-DeepNet, which combines VGG-19 and Vision Transformer (ViT) architectures for more precise classification of ALL stages. Two publicly available datasets were utilized for improved generalization: Dataset A (comprising 3,242 images) and Dataset B (comprising 2,857 images) and both datasets include four stages of ALL. The images were preprocessed through resizing, rotation, flipping, normalization, and augmentation techniques. This proposed hybrid model integrates VGG-19 and ViT with custom connected layers, batch normalization, and GeLU activation functions to generate classification results. The proposed model attained accuracies of 99.65% on dataset A and 99.69% on dataset B. To validate robustness, a 3-fold cross-validation was performed, resulting in average validation accuracies of 99.88% for dataset A and 99.91% for dataset B. Additionally, explainable AI methods were integrated into the hybrid model to enhance transparency and highlight the critical features that drive its predictions. This study addresses the crucial challenge of model interpretability and reliability, thereby contributing to more trustworthy and effective ALL stage classification.

**Index Terms**—Deep learning, CNN, Image classification, XAI, GradCam, LIME, Blood cancer

## I. INTRODUCTION

Acute Lymphoblastic Leukemia (ALL) is a severe form of blood cancer that mainly impacts children and adolescents [1]. It occurs when the bone marrow generates an overabundance of immature white blood cells, which increase uncontrollably and can spread to other areas of the body [2]. Detecting ALL accurately in the early stages is very important because it helps doctors choose the best treatment and increases the chances of recovery. Nowadays, doctors usually identify ALL by manually looking at blood samples under a microscope. This process is time-consuming, depends on the expert's experience, and can lead to mistakes, especially when there are not enough trained professionals.

Recently, Artificial Intelligence (AI) has advanced significantly with the development of deep learning. Moreover,

deep learning is widely used to analyze medical images faster and more accurately [3], [4]. However, many existing AI models still have several limitations. These include limited accuracy, poor generalization across different datasets, and a lack of transparency in decision-making, which reduces user trust and hinders their adoption in real-world clinical settings. For example, Faiz et al., [5] evaluated multiple ML models (SVM, RF), DL architectures (CNN, AlexNet, ResNet50, ShuffleNet, MobileNetV2, RNN), and hybrid combinations (MobileNetV2+SVM, MobileNetV2+RF) for leukemia detection. However, the research is constrained by very limited datasets (a total of 368 images only). Similarly, Genovese et al. [6] worked on a deep learning framework for leukaemia classification. The primary limitation is the complex model architecture, which requires substantial computational time for training and inference. Elsayed et al. [7] evaluated multiple architectures such as AlexNet, MobileViTv2, CNN Ensemble, Texture-CNN+SD, and Fuzzy+RBFNN, for leukaemia classification. The main limitation is the complex model architecture, and no explainable AI methods were incorporated for interpretability. Explainable AI aims at addressing this problem by creating more transparent systems of AI. It broadly aims at delivering methods that aid model interpretability and comprehensibility without sacrificing high performance.

This study aims to tackle major challenges in creating a more effective, innovative, and interpretable hybrid deep learning model for ALL diagnosis. We want this model not only to perform well but also to explain how it makes decisions, so that doctors can trust and understand its results. The main contributions of this study are given below:

- Developed a robust hybrid model for more accurate and precise stage classification of ALL.
- Integrated explainable AI techniques to enhance transparency in the model's decision-making.
- Used publicly available two large and diverse datasets to make sure the model performs well on various data.
- Performance was evaluated in two ways: first, by dividing the data into training, validation, and test sets; and second, by applying 3-fold cross-validation to evaluate the robustness.

## II. LITERATURE REVIEW

Recently, deep learning models have been extensively employed by researchers for the classification of acute lymphoblastic leukemia. For instance, Jawahar et al. [8] proposed a deep learning model called DDRNet for classifying ALL. A blood cell images dataset, collected from Kaggle, consists of 16,250 images used in this study. Their proposed model obtained an accuracy of 91.98%. The low testing accuracy and the limited use of XAI techniques were the limitations of their study. Next, Muduli et al. [9] explored multiple pre-trained deep learning models to identify the best-performing model for the detection and classification of acute lymphoblastic leukaemia. Their applied leukaemia dataset consists of a total of 3,242 PBS images used in their study for the classification. Among all models, DenseNet-121 performed best, achieving 99% accuracy and 98% AUC. Additionally, Papaioannou et al. [10] deployed Convolutional Neural Networks, including MobileNetV2, ResNet50, and VGG19, with customized parameters to classify leukemic B-lymphoblast cells from an image dataset comprising 10,661 blood smear images. Resizing, grayscale conversion, rotation, flipping, and normalisation are the preprocessing techniques applied to the raw images. Their proposed CNN model, ConvNet, achieved higher accuracy with 94.3%. Furthermore, Tusar et al. [11] classified ALL cells using MobileNetV2, ConvNet, and VGG19 on a dataset comprising 3,256 PBS images from patients with ALL and healthy individuals. Preprocessing steps include resizing, padding, normalisation, and augmentation. MobileNetV2 achieved an accuracy of 97.42%, outperforming other models. This study does not include XAI techniques. A summary of recent related studies is shown in Table I.

TABLE I: Summary of Recent Related Studies

| Study | Dataset                  | Used Model                        | Performance and Limitations                                                                        |
|-------|--------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------|
| [12]  | ALL-IDB2                 | Chronological Sine Cosine         | Accuracy: 81.92%. Limitation: Single dataset, low generalization and classification accuracy.      |
| [13]  | ALL-IDB1, ALL-IDB2       | CNN                               | Accuracy: 96.97%. Limitation: Insufficient dataset size, model not fully generalized.              |
| [14]  | ALL-IDB1, ALL-IDB2       | Hybrid (MobileNetV2 and ResNet18) | Accuracy: ALL-IDB1(99.39%), ALL-IDB2(97.18%). Limitation: Complex model, no decision transparency. |
| [15]  | B-ALL microscopic images | NASNetLarge + VGG19               | Accuracy: 96.58%. Limitation: Complex model, no decision transparency.                             |
| [16]  | ISBI 2019                | All-Net                           | Accuracy: 95.54%, Specificity: 95.81%. Limitation: No decision transparency.                       |

## III. METHODOLOGY

### A. Data Acquisition and Pre-processing

In this study, we used two publicly available datasets from Kaggle, referred to as Dataset A [17] and Dataset B [18]. Both datasets contain four classes. Dataset A consists of

3,242 images. Dataset B contains 2,857 images categorised as Benign (505), Early Malignant (985), Pre-Malignant (563), and Pro-Malignant (804). An overview of datasets is shown in Table II.

TABLE II: Description of the Datasets

| Serial         | Class Name              | Number of Images | Total |
|----------------|-------------------------|------------------|-------|
| Dataset A [17] |                         |                  |       |
| 1              | Benign                  | 512              | 3,242 |
| 2              | Pre-B (Malignant)       | 955              |       |
| 3              | Pro-B (Malignant)       | 796              |       |
| 4              | Early Pro-B (Malignant) | 979              |       |
| Dataset B [18] |                         |                  |       |
| 1              | Benign                  | 505              | 2,857 |
| 2              | Early-Malignant         | 985              |       |
| 3              | Pre-Malignant           | 563              |       |
| 4              | Pro-Malignant           | 804              |       |

To maintain consistency and improve model performance, all images were resized to a standard resolution of 224 × 224, normalized, and augmented using standard techniques, including random rotation (30°), horizontal and vertical flipping, and zooming. In addition, the Gaussian blur with a kernel size of (3×3) and sigma in the range of 0.1 to 2.0 was used to mimic potential imaging noise, and random grayscale conversion with a probability of 0.1. This technique was used to simulate potential monochromatic tissue appearances. Finally, the images were converted to tensors to prepare them for input into the hybrid neural network. These preprocessing steps increased the diversity of the training data and improved the generalization ability of the deep learning models. After that, both datasets were divided into training, validation, and test sets in a 60:20:20 ratio.

### B. Proposed Hybrid Model

A hybrid deep learning architecture named TM-DeepNet is proposed in this study. TM-DeepNet is a hybrid DL model that integrates VGG-19 [19] and Vision Transformer (ViT) [20] for robust classification of ALL cell cancer stages. The layer architecture of the proposed hybrid model is illustrated in Fig. 1.

1) *Feature Extraction*: VGG19 extracts local morphological features from image datasets [19], while ViT captures global contextual dependencies through self-attention. Their fusion provides a more discriminative and generalizable representation across disease stages. For an input image, VGG19 produces a flattened feature vector capturing fine-grained local patterns, and ViT generates a global embedding encoding long-range dependencies and spatial context [20]. These feature vectors provide complementary representations that are essential for accurate classification across all stages of acute lymphoblastic leukemia. Mathematically, the feature extraction process can be expressed as:

$$f_{VGG} = VGG19_{features}(x) \in \mathbb{R}^{25088} \quad (1)$$

$$f_{ViT} = ViT_{embedding}(x) \in \mathbb{R}^{768} \quad (2)$$

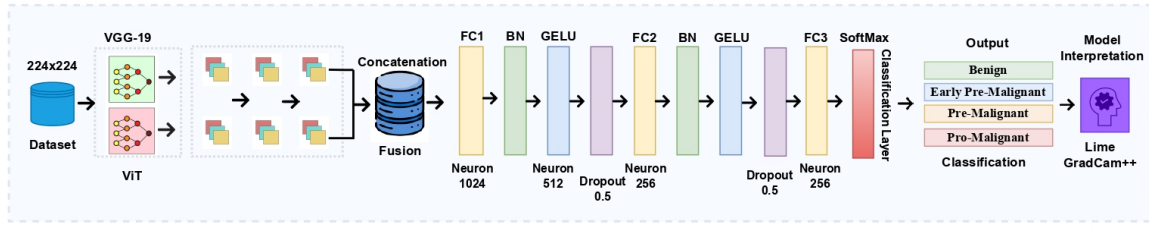


Fig. 1: Proposed TM-DeepNet Hybrid-model Layer Architecture

2) *Feature Concatenation*: The extracted features from VGG19 and ViT are concatenated to form a joint representation, can be represented as below:

$$f_{\text{concat}} = [f_{\text{VGG}} \parallel f_{\text{ViT}}] \in \mathbb{R}^{25856} \quad (3)$$

This combined representation is subsequently passed through fully connected layers incorporating batch normalization, GeLU activation, and dropout.

3) *GeLU Activation*: The GeLU activation introduces smooth non-linearity by weighting inputs according to the Gaussian cumulative distribution function [21]. It can be represented as below:

$$\text{GeLU}(z) = z \cdot \Phi(z) \quad (4)$$

$$\text{where, } \Phi(z) = \frac{1}{2} \left[ 1 + \text{erf} \left( \frac{z}{\sqrt{2}} \right) \right].$$

4) *Classification*: The hidden transformations of the fused features are defined as:

$$h_1 = \text{Dropout}(\text{GeLU}(\text{BN}(W_1 f_{\text{concat}} + b_1))) \quad (5)$$

$$h_2 = \text{Dropout}(\text{GeLU}(\text{BN}(W_2 h_1 + b_2))) \quad (6)$$

The final prediction is obtained through a SoftMax classifier:

$$\hat{y} = \text{Softmax}(W_3 h_2 + b_3) \quad (7)$$

where  $\hat{y} \in \mathbb{R}^C$  represents the probability distribution over  $C$  ALL stages. The model is trained using categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (8)$$

### C. Hyperparameter Tuning and Optimization

The proposed architecture is optimized using the Adam optimizer with a learning rate of 0.0001 and a weight decay of 0.00001. Training is conducted over 25 epochs with a batch size of 8. Dropout rates of 0.5 and 0.4 are applied to the first and second fully connected layers, respectively. A summary of the hyperparameters is provided in Table III.

By combining CNN-based (VGG-19) local features with transformer-based (ViT) global dependencies, and employing GeLU activations that provide smooth probabilistic neuron

TABLE III: Overview of Hyperparameters for TM-DeepNet

| Hyperparameter         | Value                                                 |
|------------------------|-------------------------------------------------------|
| Optimizer              | Adam                                                  |
| Learning Rate          | 0.0001                                                |
| Weight Decay           | 0.00001                                               |
| Scheduler              | StepLR (Decay factor: 0.1 every 6 epochs)             |
| Batch Size             | 8                                                     |
| Epochs                 | 25                                                    |
| Dropout (FC Layer 1)   | 0.5                                                   |
| Dropout (FC Layer 2)   | 0.4                                                   |
| Fully Connected Layers | 768 $\rightarrow$ 256 $\rightarrow$ $C$ (num classes) |
| Activation Function    | GeLU                                                  |
| Loss Function          | Categorical Cross-Entropy                             |

responses, the proposed TM-DeepNet hybrid model achieves robust and generalizable classification across all stages of acute lymphoblastic leukemia.

### D. Performance Evaluation Metrics

In this study, accuracy, precision, recall, and F1-score were used as performance measures to evaluate the classification potential of the proposed hybrid model. Furthermore, a 3-fold cross-validation was also undertaken for each dataset to validate the robustness and generalizability of the proposed model's performance. Mathematical formulations for accuracy, precision, recall, and F1-score are as follows:

$$\begin{aligned} \text{Accuracy} &= \frac{Tp + Tn}{\text{TotalSample}}, & \text{Precision} &= \frac{Tp}{Tp + Fp}, \\ \text{Recall} &= \frac{Tp}{Tp + Fn}, & \text{F1-Score} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned} \quad (9)$$

### E. Explainable AI Interpretable Techniques

1) *GradCam++*: Grad-CAM++ is a higher-order explainable AI technique for explaining and visualizing the output of deep learning models, particularly for image classification tasks [22]. It is a modification of the original Grad-CAM, providing a more accurate and comprehensive localization of regions in the input image that are accountable for the model's prediction. Although Grad-CAM employs a linear assumption for each pixel's responsibility and a global average pooling of gradients, Grad-CAM++ utilizes weighted sums of higher-order derivatives to provide a more comprehensive account of multiple regions of the same class within an image,

as well as sharper attention maps [23]. This renders Grad-CAM++ more accurate for interpreting predictions for medical imaging, object detection, and other complex vision tasks. Mathematically, it can be detailed as follows:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial^2 Y^c}{\partial A_k^2} \quad (10)$$

2) *LIME*: Lime is an explainable AI technique that allows us to understand the decisions made by machine learning models: Why was a particular decision made by the model, particularly regarding the features the model used to reach that decision [24]. This method is suitable when the model is highly complex or unclear, for example, when using hybrid deep learning or ensemble models.

#### IV. RESULTS AND DISCUSSION

##### A. Experimental Setup

The experiments were conducted using Kaggle T4 GPU. Proposed hybrid deep learning model was implemented in PyTorch (version 2.2.1) to maintain compatibility with the model requirements. Additionally, supporting libraries such as NumPy and Matplotlib were employed as part of the experimental setup.

##### B. Performance Analysis of the Proposed Model

The proposed hybrid deep learning model demonstrated excellent performance in classifying all four stages of ALL. It achieved training accuracies of 99.74% on dataset A and 99.80% on dataset B after 25 epochs of training. The detailed performance metrics for training, testing, and validation on both datasets are presented in Table IV. 3-fold cross-validation results are summarized in Table V. These results highlight the model's strong ability to learn and represent complex patterns within the datasets. Its generalizability was further validated by average test accuracies of 99.65% on dataset A and 99.82% on dataset B, indicating highly reliable predictive performance on unseen data. Moreover, the consistent decrease in training and validation losses throughout the process reflects steady model improvement without signs of overfitting.

TABLE IV: Performance Metrics Summary

| Phase            | Accuracy (%) | Precision (%) | Recall (%) | F1-score (%) |
|------------------|--------------|---------------|------------|--------------|
| <b>Dataset A</b> |              |               |            |              |
| Training         | 99.74        | 99.74         | 99.68      | 99.71        |
| Validation       | 99.54        | 99.33         | 99.62      | 99.47        |
| Testing          | 99.65        | 99.60         | 99.66      | 99.63        |
| <b>Dataset B</b> |              |               |            |              |
| Training         | 99.80        | 99.73         | 99.79      | 99.76        |
| Validation       | 99.85        | 99.73         | 99.87      | 99.80        |
| Testing          | 99.69        | 99.75         | 99.60      | 99.67        |

TABLE V: 3-Fold Cross-Validation Accuracy(%)

| Dataset   | Fold 01 | Fold 02 | Fold 03 | Average |
|-----------|---------|---------|---------|---------|
| Dataset A | 99.81   | 100.00  | 99.81   | 99.88   |
| Dataset B | 99.82   | 99.91   | 100.00  | 99.91   |

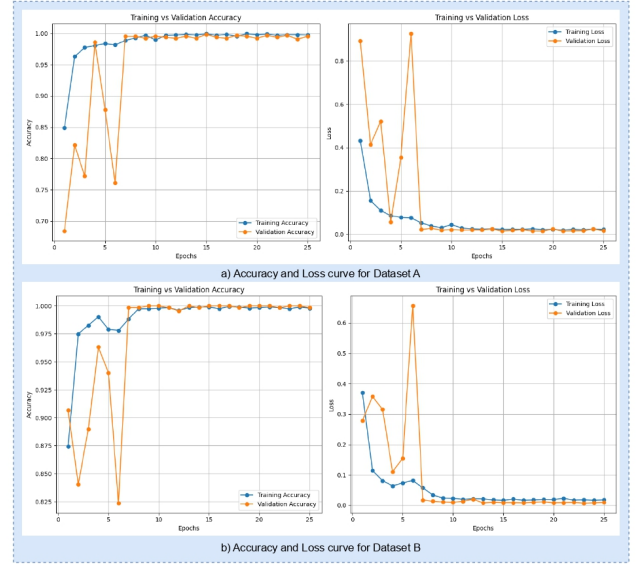


Fig. 2: Accuracy and Loss Curve Analysis

##### C. Accuracy and Loss curve analysis

The accuracy curve represents the model's classification performance across epochs, while the loss curve reflects variations in the model's error during training. Fig. 2 presents the training and validation accuracy and loss curves for both datasets. The results show that the loss decreases significantly after four epochs and stabilizes around the seventh epoch. Similarly, the accuracy improves notably after four epochs and reaches stability after approximately eight epochs. The training and validation accuracy curves follow a nearly identical trajectory without any significant gap, while the validation loss remains steady as the training loss decreases, indicating the absence of overfitting. Furthermore, both training and validation losses start at reasonable values and decline rapidly, confirming that the model learns effectively without signs of underfitting.

##### D. Confusion Matrix Analysis

Fig. 3 presents the confusion matrices for the multiclass classification of the four stages of ALL using the proposed TM-DeepNet model. Diagonal entries correspond to correctly classified samples, while off-diagonal entries indicate misclassifications. For dataset A (Fig. 3a), the model achieved near-perfect classification: 96 Benign, 193 Pre-B, 159 Pro-B, and 198 Early Pre-B samples were correctly classified, with only minor misclassifications (1 Pre-B predicted as Benign, 1 Pre-B as Pro-B, and 1 Early Pre-B as Benign). For dataset B (Fig. 3b), the model also demonstrated excellent performance: 93 Benign, 197 Early, 203 Pre, and 158 Pro samples were correctly classified. Only a single Early-stage sample was misclassified as Benign, while all other classes showed perfect classification. These results highlight the model's robustness in distinguishing between different stages of ALL with minimal

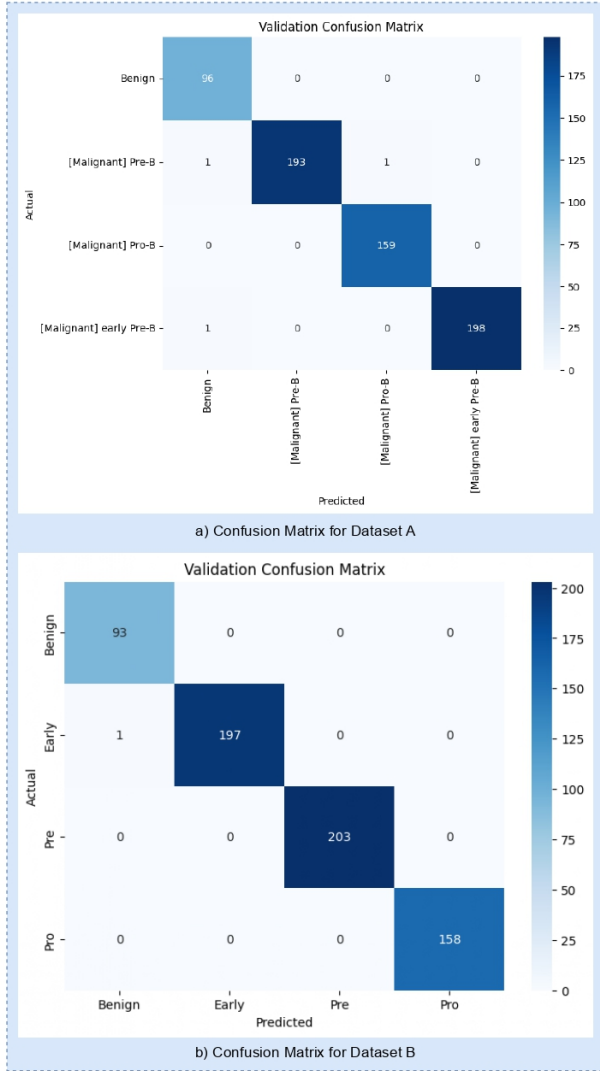


Fig. 3: Confusion Matrix Analysis

misclassification, demonstrating its strong predictive reliability.

#### E. Model Interpretability Analysis

In this study, LIME and Grad-CAM++ were employed to explain the decision-making process of the proposed classification model. As shown in Fig. 4, the first column displays the original cell images across four classes. The second column shows superpixel segmentation, where images are divided into interpretable regions. The third column highlights the most important superpixels selected by LIME, marked in yellow, which contribute most to the prediction. The last column presents Grad-CAM++ heatmaps, where red areas indicate the regions most influential in the model's decision-making.

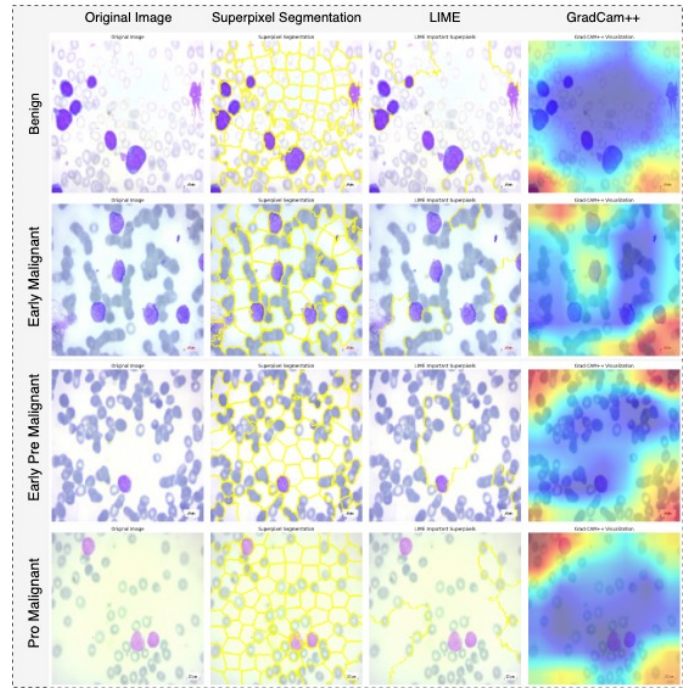


Fig. 4: Model Interpretability Analysis

#### F. Discussion

This study proposes a hybrid deep learning model that integrates VGG-19 and ViT to classify the four stages of ALL. The proposed model achieved 99.65% accuracy on dataset A and 99.69% accuracy on dataset B, demonstrating its robustness in classifying all stages of ALL. Compared to recent state-of-the-art models according to Table VI, this model outperformed existing proposed models, such as CNN and ResNet101. MobileNetV2, VGG 16 and AlexNet. As medical AI models should be explainable, XAI techniques such as GradCAM++ and Lime are integrated to enhance explainability and highlight the importance of regions in the image that impact model decisions. Applying the proposed hybrid model provides complementary feature extraction, improved accuracy, better generalization and flexibility in feature fusion, allowing it to classify different ALL stages effectively. Additionally, medical experts can examine the highlighted regions produced by LIME and Grad-CAM++ and confirmed that these areas correspond to diagnostically significant features in ALL. This evaluation demonstrates that the model's explanations are medically meaningful and can support real-world clinical decision-making.

TABLE VI: Comparison with Recent Studies

| Study                   | Model Used         | Accuracy (%)                | XAI        |
|-------------------------|--------------------|-----------------------------|------------|
| Chen et al. [25]        | Ensemble ResNet-10 | 84.09                       | No         |
| Anand et al. [26]       | CNN                | 96.00                       | No         |
| Abd El-Aziz et al. [27] | EfficientNetV2-S   | 97.33                       | No         |
| Sriram et al. [28]      | VGG-16             | 98.40                       | No         |
| Shafique et al. [29]    | AlexNet            | 92.00                       | No         |
| <b>Proposed Model</b>   | <b>TM-DeepNet</b>  | <b>99.65 (A), 99.69 (B)</b> | <b>Yes</b> |

1) *Limitations*: The limitations of the paper include VGG-19 and ViT, both of which are complex and more expensive than the other models deployed for the same classification task. As the proposed model is a hybrid of two complex models, integration difficulties may be faced during implementation. Larger or more complex models, such as the proposed model, are harder to integrate into edge devices.

## V. CONCLUSION AND FUTURE WORKS

This paper proposed a hybrid deep learning model, TM-DeepNet, for more accurate classification of Acute Lymphoblastic Leukemia (ALL) stages. The model integrates VGG-19 and Vision Transformer (ViT) architectures with custom connector layers, batch normalization, and GeLU activation functions to generate precise classification results. The proposed model achieved an accuracy of 99.65% on dataset A and 99.69% on dataset B, demonstrating its robust performance across the four stages of ALL. Furthermore, the integration of explainable AI techniques, such as Grad-CAM++ and LIME, enhances the model's trustworthiness, transparency, and interpretability by highlighting the decision-making process in a manner that is human-understandable.

For future work, lightweight CNN-based models can be explored to improve computational efficiency further. Additionally, model compression techniques such as pruning, quantization, and knowledge distillation may be applied to facilitate deployment on edge devices, making the system more practical for real-world clinical applications.

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